

Steve Broll

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Research interests

High-dimensional models, variable selection, multi-omics data, time-varying outcomes, wearable data, data fusion, data visualization, tensor decomposition and regression

Education

- 2020 – 2025 **Cornell University** – Ithaca, New York
PhD in Statistics
Dissertation: *Variable Selection for High-Dimensional Longitudinal Omics Data with a Continuous or Misclassified Binary Outcome*
Committee: Martin T. Wells, Sumanta Basu, Myung Hee Lee
- SMoCS Lab Member (Statistical Modeling of Complex Systems)
Predoctoral Fellow in AI and Precision Nutrition
- 2020 – 2023 **Cornell University** – Ithaca, New York
MS in Statistics
- 2016 – 2020 **Texas A&M University** – College Station, Texas
BS in Statistics
Minor in Mathematics

Awards and Fellowships

- 2024 AI and Precision Nutrition (AIPrN) NIH T32 Predoctoral Fellow
- 2024 Cornell Graduate School Conference Travel Grant
- 2023 Cornell Graduate School Conference Travel Grant
- 2020 Dr. Newton Service Award (Texas A&M University)
- 2018 Statistics Department Scholarship (Texas A&M University)
- 2018 First Prize, Texas A&M University Institute of Data Science Undergraduate Competition
- 2016 President's Endowed Scholarship (Texas A&M University)

Publications

- 2025 **Variable Selection for High-Dimensional Longitudinal Omics Data with a Continuous or Misclassified Binary Outcome**
Steven Broll.
Cornell University.
- 2025 **PROLONG: Penalized Regression for Outcome guided Longitudinal Omics analysis with Network and Group constraints**
Steven Broll, Sumanta Basu, Myung Hee Lee, Martin T Wells.
Bioinformatics.
- 2021 **Interpreting blood GLUcose data with R package iglu**
Steven Broll, Jacek Urbanek, David Buchanan, Elizabeth Chun, John Muschelli, Naresh M Punjabi, Irina Gaynanova.
PLOS One.

Preprints

- 2025 **Urine metabolomic biomarkers linked to C-reactive protein-interleukin-6 axis in persons living with HIV and tuberculosis**
Andrea B. Doltrario, Myung Hee Lee, **Steven Broll**, Kathryn Dupnik, Vanessa Rouzier, Patrice Severe, Nancy Dorvil, Jean W Pape, Serena P. Koenig, Daniel W. Fitzgerald, Kyu Y. Rhee.
medRxiv.

Research experience

- 2026 – Present **Michigan Data Science Fellow**
Advisors: Irina Gaynanova, Sung Kyun Park
Variable selection, network analysis, and data fusion for multi-omic, environmental exposure, and wearable data with applications to type-2 diabetes.
- 2025 – 2026 **Postdoctoral Associate**
Advisors: Julia Finkelstein, Saurabh Mehta, Martin T. Wells
Continued work on high-dimensional variable selection with emphasis on applications to precision nutrition and health data. Statistical analysis and replication of various nutrition studies.

- 2024 – 2025 **T32 Predoctoral Fellow**
 Committee: Martin T. Wells, Sumanta Basu, Myung Hee Lee, Saurabh Mehta
 Continued work on extensions to PROLONG, with applications to omics and precision nutrition data. Debiased group lasso and inference for treatment interaction modeling. EM algorithm for binary outcome with potential misclassification.
- 2021 – 2023 **Graduate Research Assistant**
 Committee: Martin T. Wells, Sumanta Basu, Myung Hee Lee
 Batch effect correction, high-dimensional network modeling and differential expression analysis for proteomic data. Batch effect correction and high-dimensional modeling for metabolomic data from various *Mtb* cohorts. Developed PROLONG model and worked on various extensions.

Teaching experience

- Spring 2025 **Teaching Fellow, STSCI 2000: Essential Statistics and Data Science , eCornell - National Ed Equity Lab (NEEL) Partnership**
- Fall 2020 **Teaching assistant, BTRY 6010: Statistical Methods 1, Cornell University**

Talks and Posters

Variable Selection for High-Dimensional Longitudinal Omics Data with a Continuous or Misclassified Binary Outcome

- July 2025 Cornell University B-Exam
- Uncertainty Quantification and Inference with PROLONG**
- April 2025 Cornell University Center for Precision Nutrition and Health
- PROLONG: Penalized Regression for Outcome guided Longitudinal Omics analysis with Network and Group constraints**
- November 2024 Cornell University Center for Precision Nutrition and Health
- August 2024 Joint Statistical Meetings
- March 2024 International Biometric Society Eastern North American Region (ENAR) Meeting
- September 2023 Cornell Celebration of Statistics and Data Science (Poster)
- August 2023 Cornell University A-Exam

August 2023	Joint Statistical Meetings
March 2023	International Biometric Society Eastern North American Region (ENAR) Meeting

Mentorship and service

April 2025 – Present	eCornell Course Development Assisted in development of a course on using R, SQL, and the All of Us researcher workbench for precision nutrition research.
February 2021 – December 2022	Cornell Directed Reading Program Met weekly with an undergraduate student each semester, selecting a book or set of papers of mutual interest, covering topics from spatiotemporal modeling, statistical learning, measure theoretic probability, nonparametric regression and GAMs.
September 2017 – May 2020	Statistics Peer Mentor Developed and maintained relationships with and provide assistance and support for first-year students (focus on a smooth transition, acclimation, and a sense of belonging). Maintained regular contact with five to six assigned students, and served as a positive academic and social role model.

Technical skills

Programming languages

Proficient in: R, Shiny, Quarto

Familiar with: Python, C++, Stan

Software

LaTeX, Git

Selected Coursework

Distribution Theory, Theory of Inference, Theory of Linear Models, Multivariate Analysis, Flexible Regression using R, Statistical Computations, Computationally Intensive Stat Methods I and II, Spatial Data Analysis, Time Series and Spatiotemporal Data Science, Mathematical Statistics I and II, Generalized Linear Models, Advanced Statistical Consulting, Bayesian Statistics and Data Analysis, Principles of Analysis II, Probability I and II, Deep Learning Theory and Applications.